

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Concept Mapping (Annexure A)

Course Name: Cryptography and Network Security

Course Code: CS A5117

Course Outcome CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security. (BL3, BL4)

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Question 1 - Security Attacks, Threats, Risks and Security Goals

Concept map linking Security Attacks, Threats, Risks and the Security Goals (Confidentiality, Integrity, Availability), showing how each attack leads to specific threats and risks, how the goals mitigate them, and one real-world example per link.

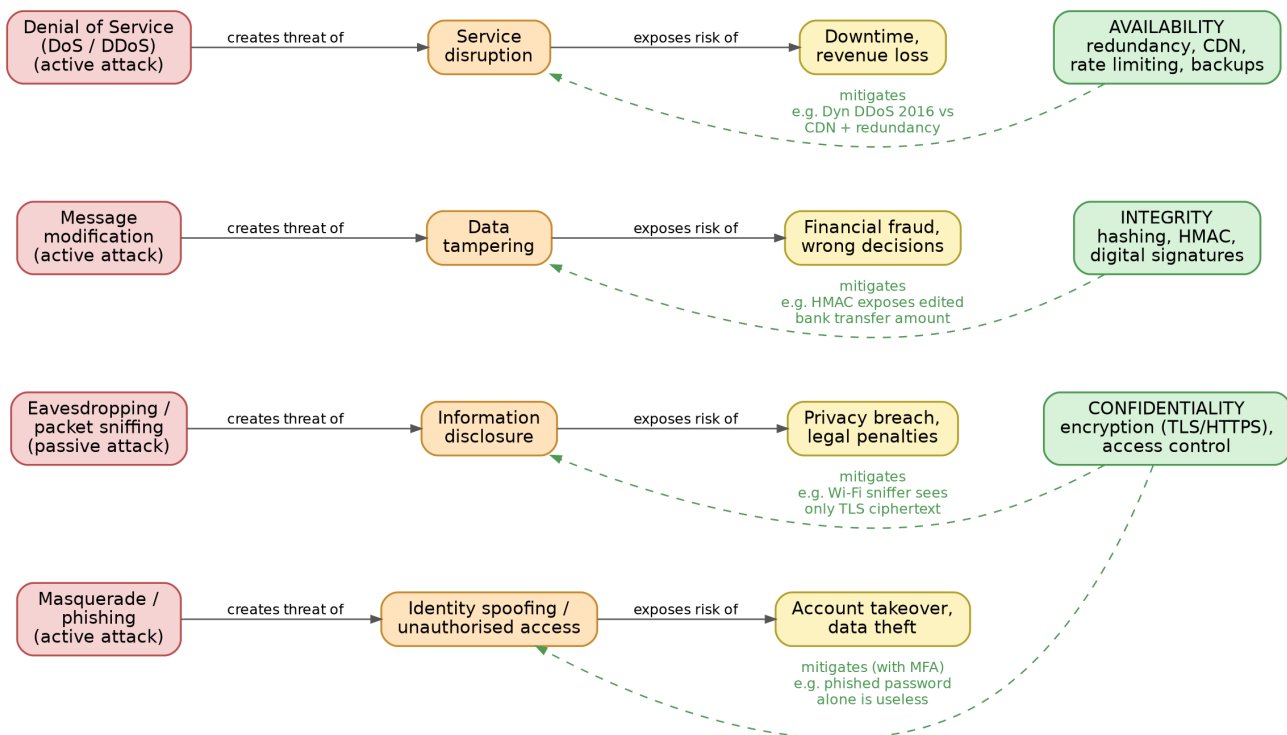


Fig. 1 - Attack → Threat → Risk chains, with the CIA goals cutting each chain (dashed green links = mitigation)

How the map reads

I have arranged the map as a chain that flows from left to right. An attacker performs some action (the attack), that action creates a threat, and the threat exposes the organisation to an actual risk - something measurable like money lost, data leaked or downtime. The three security goals sit on the right and push back against these chains, which is why their links point backwards into the threats.

(a) How each attack leads to specific threats and risks

Eavesdropping → information disclosure → privacy breach. A passive attacker just listens. Nothing is altered, so the victim may never even know, but confidential data leaks out and legal penalties (say, under data protection law) follow. *Example:* packet sniffing on open airport Wi-Fi.

Message modification → data tampering → financial fraud. An active attacker intercepts a message and changes it in transit. If the amount or account number in a bank transfer gets edited, the bank ends up making a wrong payment. *Example:* a man-in-the-middle altering a transaction request.

Masquerade / phishing → identity spoofing → account takeover. The attacker pretends to be someone trusted, gains unauthorised access, and from there steals data or money. *Example:* a fake 'bank' e-mail that harvests login credentials.

DoS / DDoS → service disruption → downtime and revenue loss. Flooding a server does not steal anything, but legitimate users are locked out. *Example:* the 2016 Dyn DDoS

attack that knocked Twitter, Netflix and Reddit offline for hours.

(b) How the security goals mitigate them

Confidentiality cuts the eavesdropping chain: once traffic runs over TLS/HTTPS, the sniffer on that airport Wi-Fi only captures ciphertext. Combined with multi-factor authentication it also blunts phishing, because a stolen password alone is no longer enough to get in. **Integrity** cuts the tampering chain: an HMAC or digital signature on the bank message means any edited bit changes the tag, so the modification is detected before the payment happens. **Availability** cuts the DoS chain: redundant servers, rate limiting and CDNs absorb or deflect the flood, which is exactly how large sites recovered after the Dyn incident.

Analysis

One pattern that stands out from the map: passive attacks mainly threaten confidentiality and are hard to detect, so for them *prevention* (encryption) matters most. Active attacks threaten integrity and availability and are easier to notice, so *detection and recovery* carry more weight there. Also, the goals are not isolated - a single strong control like MFA sits on the confidentiality side but ends up cutting the spoofing chain too, so in practice one good control can break several attack chains at once.

Question 2 - Comparing Substitution Techniques

Concept map comparing Caesar, Monoalphabetic, Vigenère and Hill ciphers on key usage, complexity and security strength, with an analysis of which technique is most secure and why.

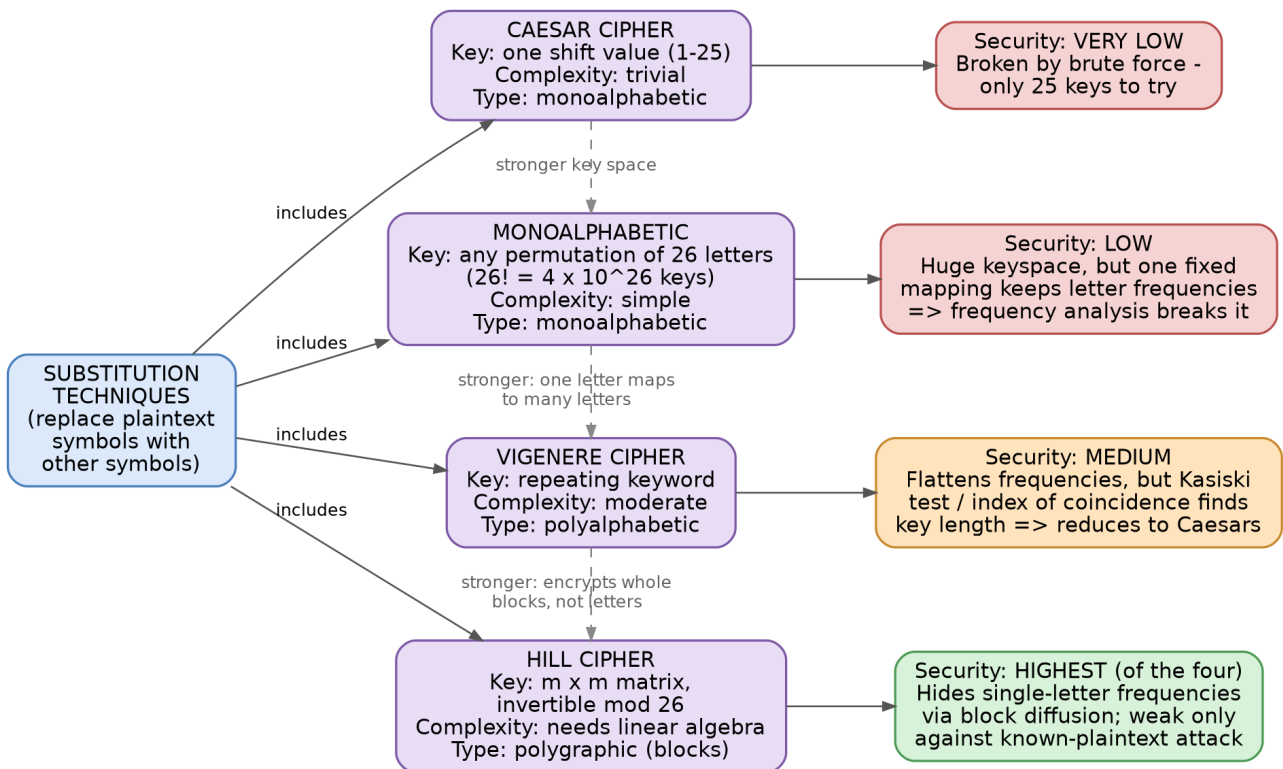


Fig. 2 - The four substitution ciphers with their keys, and how each one gets broken

Key usage and complexity

The **Caesar cipher** uses a single shift value between 1 and 25 - about as simple as a key can get. The **monoalphabetic cipher** replaces the shift with an arbitrary permutation of the alphabet, which explodes the keyspace to $26!$ (roughly 4×10^{26} keys) while staying easy to operate by hand. The **Vigenère cipher** moves to a repeating keyword: each plaintext letter is shifted by a different amount depending on its position, which is what makes it polyalphabetic. The **Hill cipher** is the odd one out - its key is an $m \times m$ matrix that must be invertible mod 26, and encryption is matrix multiplication on blocks of m letters, so it actually needs some linear algebra to run.

Security strength (why each one falls)

Caesar falls to plain brute force - 25 candidate keys can be tried in seconds, even on paper. The monoalphabetic cipher looks impressive with its $26!$ keyspace, but the mapping is fixed, so every 'e' in the plaintext becomes the same ciphertext letter. English letter statistics leak straight through, and frequency analysis breaks it without ever brute-forcing the key. That is an important lesson from this comparison: **a huge keyspace is worthless if the cipher leaks structure**. Vigenère flattens those single-letter frequencies, which kept it 'unbreakable' for about three centuries, but the repeating keyword is its weakness - the Kasiski examination or the index of coincidence reveals the key length, and after that the ciphertext splits into several independent Caesar ciphers. Hill destroys single-letter frequencies altogether, because each ciphertext letter depends on m plaintext letters at once. Its real weakness is the known-plaintext attack: since encryption is linear, a few plaintext-ciphertext block pairs give a system of linear equations that recovers the key

matrix.

Analysis - which is most secure and why

Ranking them: **Caesar < Monoalphabetic < Vigenère < Hill**. I would place the Hill cipher on top because it is the only polygraphic cipher of the four - it diffuses every plaintext letter across a whole block, so the frequency tricks that kill the other three simply do not apply to it. Its known-plaintext weakness also demands more from the attacker: they must already possess matching plaintext-ciphertext pairs, which is a stronger assumption than merely having intercepted ciphertext. Vigenère deserves an honourable mention, though - if the keyword were truly random and as long as the message, it would become the one-time pad, which is provably unbreakable. But with a normal short repeating keyword, which is how it is actually used, Hill comes out clearly ahead.

Question 3 - Substitution vs Transposition Techniques

Concept map for classical encryption schemes: differences and similarities between substitution and transposition, their effect on data structure and security level, and an analysis of which is more vulnerable to attacks and why.

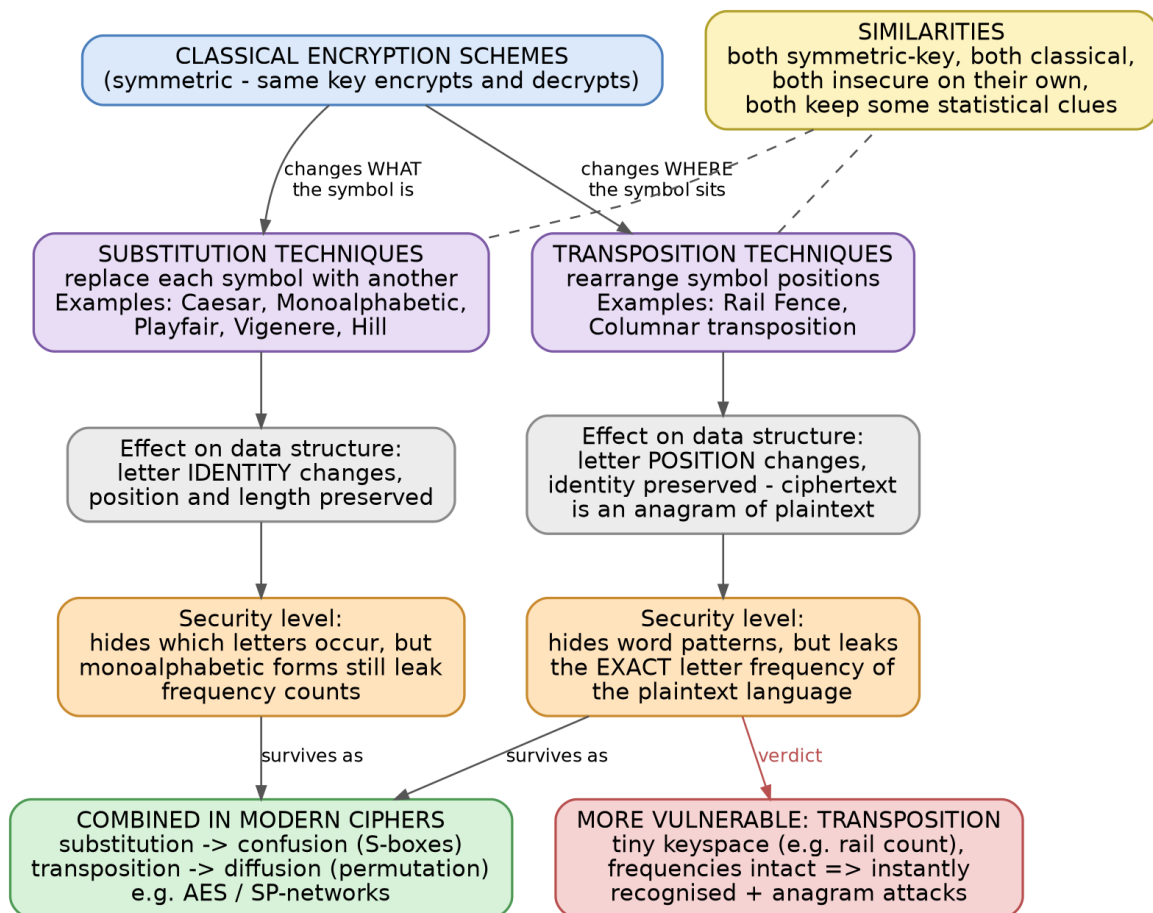


Fig. 3 - The two families of classical encryption and where each one leaks

(a) Differences and similarities

The core difference is easy to state: substitution changes **what** a symbol is, transposition changes **where** it sits. In a substitution cipher every letter's identity is replaced but its position survives; in a transposition cipher every original letter is still present, just shuffled - the ciphertext is literally an anagram of the plaintext. The similarities matter just as much: both are symmetric (the same key encrypts and decrypts), both are classical pen-and-paper era schemes, both are insecure on their own, and both leave statistical fingerprints behind for a cryptanalyst to grab.

(i) Effect on data structure and security level

Substitution preserves message length and letter order while altering symbol values, so its security depends on how well those values are disguised - a monoalphabetic scheme still leaks frequency counts, while polyalphabetic ones smooth them out. Transposition preserves the exact multiset of letters while permuting order, so its security depends entirely on hiding the permutation - and it can never hide the fact that the ciphertext has the letter-frequency profile of ordinary English. Interestingly, neither idea died: modern block ciphers like AES combine both, using substitution boxes for *confusion* and permutation layers for *diffusion*. The classical weaknesses only exist because each technique was used alone.

(ii) Analysis - which is more vulnerable and why

In my assessment, **transposition is the more vulnerable family**, at least in its simple classical forms, for three reasons. First, the key space is tiny - the Rail Fence cipher's key is just the number of rails, so brute force finishes instantly. Second, it announces itself: because ciphertext frequencies exactly match normal English, the attacker knows within seconds that a transposition was used and can go straight to anagramming with digram/trigram statistics. Third, partial progress compounds - once a few columns are aligned correctly, readable fragments appear and guide the rest of the attack. A monoalphabetic substitution at least forces the attacker to sit down and do proper frequency analysis. To be fair, neither survives a serious cryptanalyst for long; transposition just gives away more, faster.

Question 4 - Steganography: Data Hiding, Watermarking, Survivability

Concept map connecting data hiding, watermarking and survivability - their relationships, their contribution to secure communication, and one real-world application for each.

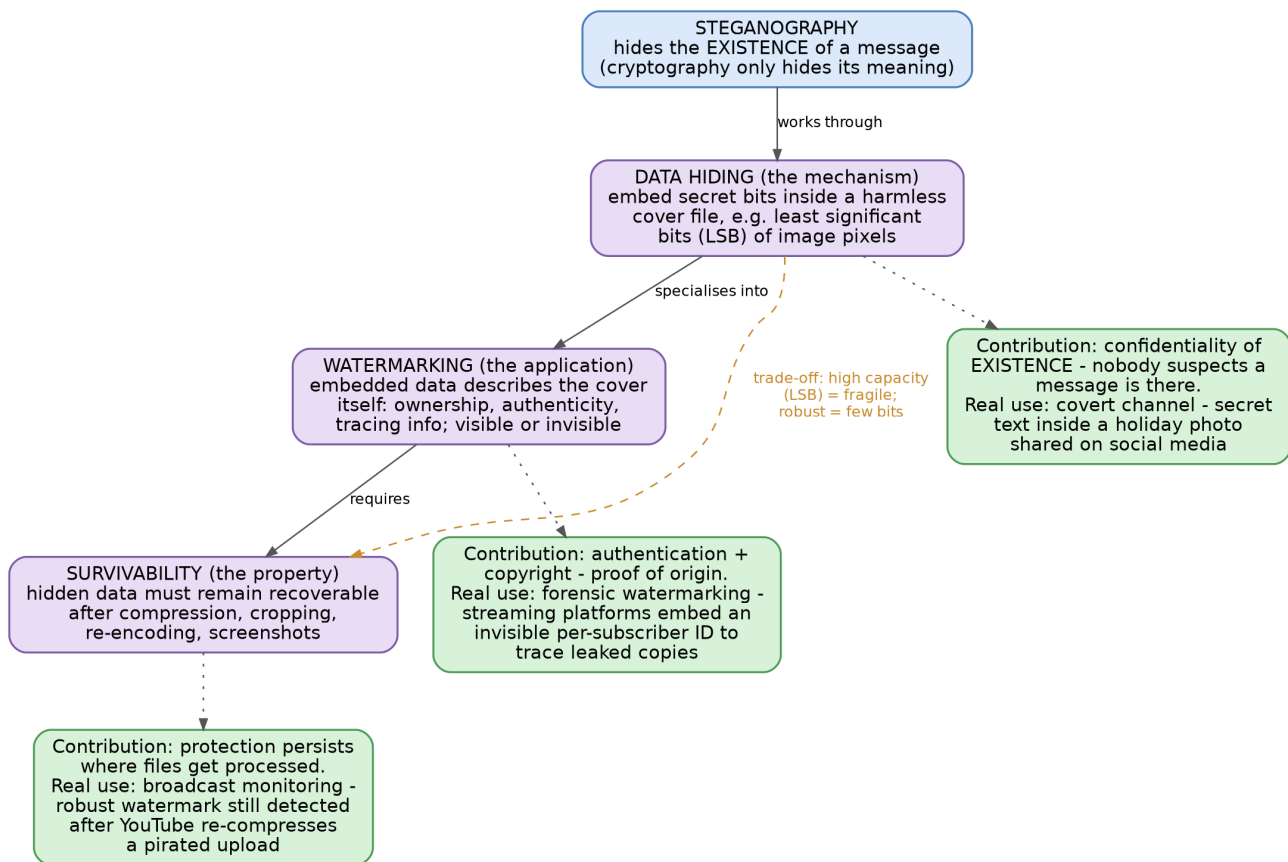


Fig. 4 - Mechanism, application and property: how the three steganography concepts depend on each other

(a) Relationships between the concepts

Where cryptography scrambles a message's meaning, steganography hides the fact that a message exists at all. Within that idea the three concepts play different roles. **Data hiding** is the underlying mechanism - embedding secret bits inside a harmless-looking cover file, for instance in the least significant bits (LSB) of image pixels, where the change is invisible to the eye. **Watermarking** is a specialised application of that mechanism: here the embedded data is *about the cover itself* - ownership, authenticity or tracing information - rather than a secret message meant for a third party. **Survivability** (robustness) is not a technique but a property the other two must satisfy: can the hidden data still be recovered after the file is compressed, cropped, converted or screenshotted? The map also shows the trade-off that ties them together: high-capacity invisible hiding like LSB is fragile and dies on the first JPEG re-save, while robust watermarks survive heavy processing but can only carry a few bits. Fragile marks are even used deliberately for tamper detection - if the mark breaks, someone edited the image.

(b) Contribution to secure communication

Each concept covers a different security need. Data hiding provides a covert channel - confidentiality of *existence*, which matters when even the act of communicating must stay hidden. Watermarking provides authentication and copyright protection, effectively a proof of

origin and a lightweight integrity check bound to the media itself. Survivability is what makes both of these useful outside the lab: real files get re-encoded, resized and compressed constantly, and a protection that vanishes on the first WhatsApp forward protects nothing.

(c) Real-world applications

Data hiding: a covert channel where a short text is embedded in the LSBs of an ordinary holiday photo posted on social media - the picture looks untouched to everyone else.

Watermarking: forensic watermarking on streaming platforms - each subscriber's video stream carries a unique invisible ID, so when pre-release content leaks, the copy can be traced back to the exact account. **Survivability:** broadcast monitoring and copyright detection - a robust watermark is still detectable after a pirated upload has been through YouTube's re-compression, cropping and format conversion, which is precisely the scenario the mark was designed to survive.

Question 5 - Number Theory Concepts in Cryptography

Concept map linking Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem - their mathematical connections, their role in encryption/decryption, and which concept is most critical for public-key cryptography.

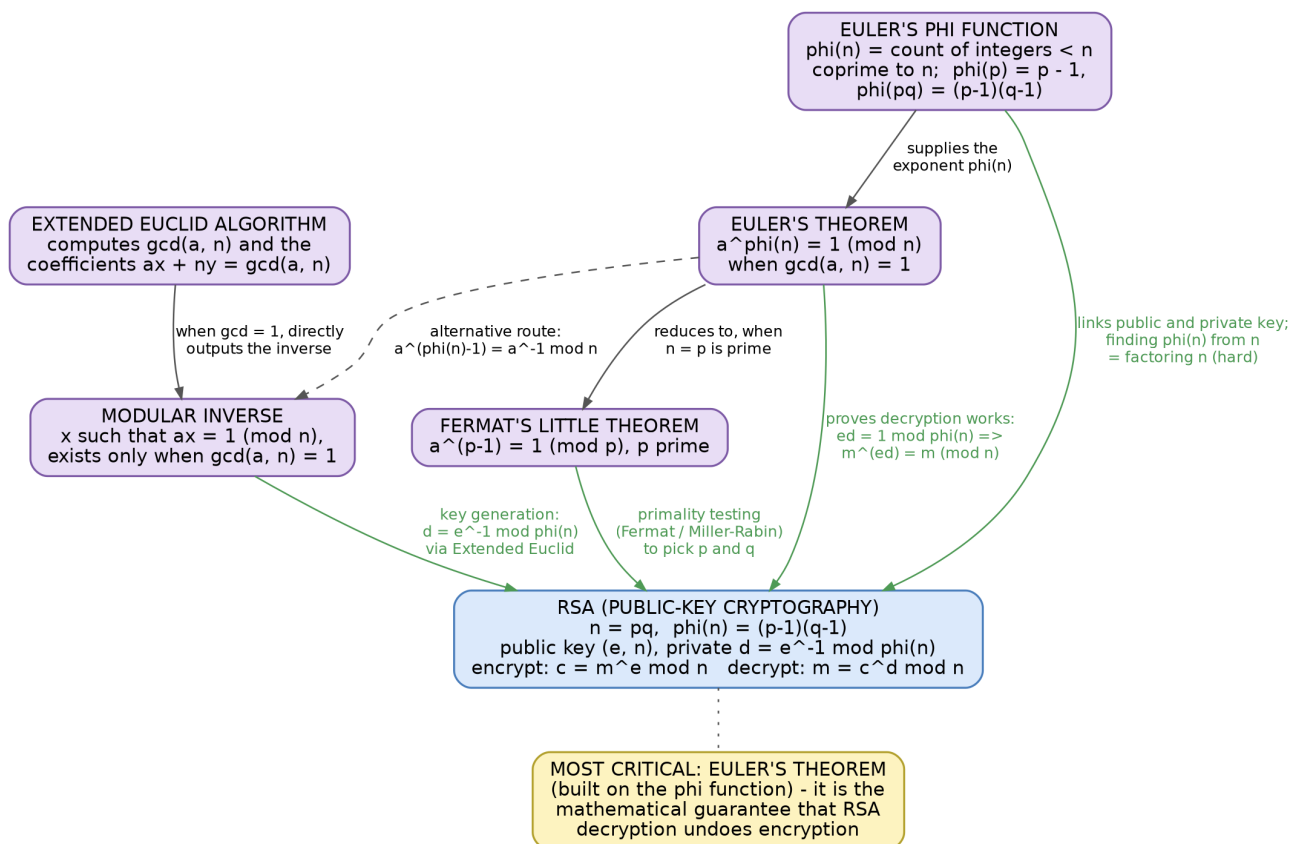


Fig. 5 - The five number theory concepts converging on RSA (green links = role in encryption/decryption)

(a) How the concepts are mathematically connected

The **Extended Euclid algorithm** is the workhorse: it computes $\gcd(a, n)$ along with coefficients x, y satisfying $ax + ny = \gcd(a, n)$. Whenever that $\gcd$ equals 1, the coefficient x is the **modular inverse** of $a \bmod n$ - the value with $ax \equiv 1 \pmod{n}$ - so the inverse exists exactly when a and n are coprime. **Euler's phi function** $\phi(n)$ counts the integers below n that are coprime to n ; for a prime p it is simply $p - 1$, and for $n = pq$ it is $(p-1)(q-1)$. **Euler's theorem** then states that $a^{\phi(n)} \equiv 1 \pmod{n}$ whenever $\gcd(a, n) = 1$, and **Fermat's little theorem** is nothing more than Euler's theorem in the special case where n is a prime p : $a^{p-1} \equiv 1 \pmod{p}$. The concepts even loop back on each other - Euler's theorem gives a second route to the modular inverse, since $a^{\phi(n)-1} \equiv a^{-1} \pmod{n}$.

(b) Their role in encryption/decryption (RSA walkthrough)

RSA uses every one of these pieces. Key generation starts by picking two large primes p and q , and this is where Fermat's little theorem quietly does its job - the Fermat and Miller-Rabin primality tests that screen the candidates are built on it. Then $n = pq$ and $\phi(n) = (p-1)(q-1)$. A public exponent e is chosen coprime to $\phi(n)$, and the private exponent is its modular inverse, $d = e^{-1} \bmod \phi(n)$, computed with the Extended Euclid algorithm. Encryption is $c = m^e \bmod n$ and decryption is $m = c^d \bmod n$. The reason decryption actually returns the original message is Euler's theorem: since $ed \equiv 1 \pmod{\phi(n)}$, we get $m^{ed} = m \cdot (m^{\phi(n)})^k \equiv m \pmod{n}$.

(c) Analysis - the most critical concept, with justification

My pick is **Euler's theorem**, together with the phi function it is built on. The justification is twofold. First, correctness: Euler's theorem is the mathematical guarantee that RSA decryption undoes encryption at all - without it, the relation $ed \equiv 1 \pmod{\phi(n)}$ would mean nothing. Second, security: RSA's safety rests on the very same object, because computing $\phi(n)$ from n is as hard as factoring n , and that hardness is exactly what keeps the private exponent d out of an attacker's reach. The other concepts are certainly important - Extended Euclid is the most useful *tool* in daily practice and Fermat's theorem makes prime generation feasible - but they exist to serve the relationship that Euler's theorem establishes. Take Euler's theorem away and RSA has neither a proof that it works nor a reason to be secure; that makes it the linchpin of public-key cryptography among these five.